

# CO1-AT1

## 1. Employee Salary Analysis

```
import pandas as pd
```

```
# Read the CSV file
```

```
df = pd.read_csv(r"C:\Users\subha\Desktop\Query processing\employees.csv")
```

```
# Display the complete employee details
```

```
print("Employee Details:\n")
```

```
print(df)
```

```
# Display employees earning more than ₹50,000
```

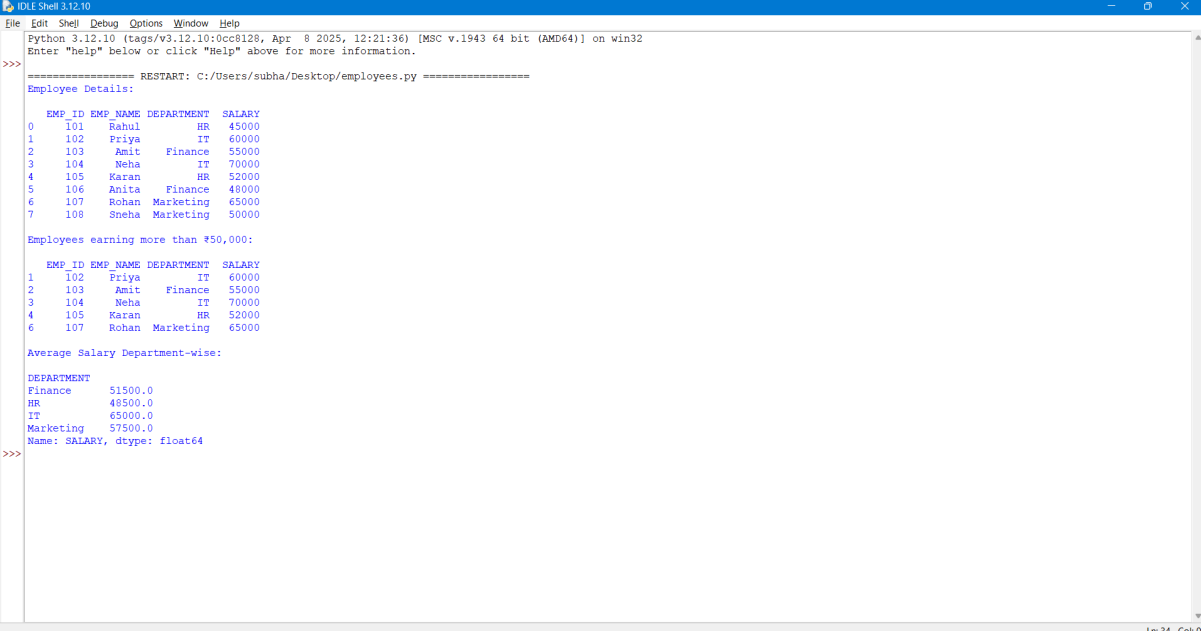
```
print("\nEmployees earning more than ₹50,000:\n")
```

```
print(df[df["SALARY"] > 50000])
```

```
# Calculate average salary department-wise
```

```
print("\nAverage Salary Department-wise:\n")
```

```
print(df.groupby("DEPARTMENT")["SALARY"].mean())
```



```
IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/subha/Desktop/employees.py =====
Employee Details:
  EMP_ID EMP_NAME DEPARTMENT  SALARY
0    101    Rahul         HR    45000
1    102    Priya         IT    60000
2    103     Amit    Finance    55000
3    104     Neha         IT    70000
4    105     Karan         HR    52000
5    106     Anita    Finance    48000
6    107     Rohan    Marketing    65000
7    108     Sneha    Marketing    50000

Employees earning more than ₹50,000:
  EMP_ID EMP_NAME DEPARTMENT  SALARY
1    102    Priya         IT    60000
2    103     Amit    Finance    55000
3    104     Neha         IT    70000
4    105     Karan         HR    52000
6    107     Rohan    Marketing    65000

Average Salary Department-wise:
DEPARTMENT
Finance    51500.0
HR         48500.0
IT         65000.0
Marketing  57500.0
Name: SALARY, dtype: float64
>>>
```

## 2. Missing Data Handling

```
import pandas as pd
```

```
# Read the CSV file
```

```
df = pd.read_csv("student_marks.csv")
```

```
# Display original dataset
```

```
print("Original Dataset:\n")
```

```
print(df)
```

```
# Identify missing values
```

```
print("\nMissing Values:\n")
```

```
print(df.isnull())
```

```
# Replace missing values with column average
```

```
df.fillna(df.select_dtypes(include='number').mean(), inplace=True)
```

```
# Display cleaned dataset
```

```
print("\nCleaned Dataset:\n")
```

```
print(df)
```

```
===== RESTART: C:/Users/subha/Desktop/Query processing/student.py =====
```

```
Original Dataset:
```

|   | Name  | Maths | Science | English |
|---|-------|-------|---------|---------|
| 0 | Ravi  | 85.0  | 90.0    | 78.0    |
| 1 | Priya | 92.0  | NaN     | 88.0    |
| 2 | Amit  | 76.0  | 81.0    | NaN     |
| 3 | Neha  | NaN   | 89.0    | 91.0    |
| 4 | Kiran | 80.0  | 85.0    | 84.0    |

```
Missing Values:
```

|   | Name  | Maths | Science | English |
|---|-------|-------|---------|---------|
| 0 | False | False | False   | False   |
| 1 | False | False | True    | False   |
| 2 | False | False | False   | True    |
| 3 | False | True  | False   | False   |
| 4 | False | False | False   | False   |

```
Cleaned Dataset:
```

|   | Name  | Maths | Science | English |
|---|-------|-------|---------|---------|
| 0 | Ravi  | 85.00 | 90.00   | 78.00   |
| 1 | Priya | 92.00 | 86.25   | 88.00   |
| 2 | Amit  | 76.00 | 81.00   | 85.25   |
| 3 | Neha  | 83.25 | 89.00   | 91.00   |
| 4 | Kiran | 80.00 | 85.00   | 84.00   |

### 3. Duplicate Record Removal

```
import pandas as pd

# Read the CSV file
df = pd.read_csv("customer_data.csv")

# Display original dataset
print("Original Dataset:\n")
print(df)

# Find duplicate records
duplicates = df[df.duplicated()]

print("\nDuplicate Records:\n")
print(duplicates)

# Remove duplicate records
df = df.drop_duplicates()

# Display updated dataset
print("\nUpdated Dataset (Duplicates Removed):\n")
print(df)
```

```
===== RESTART: C:/Users/subha/Desktop/Query processing/customer.py =====
Original Dataset:

   CustomerID  Name      City
0          101  Ravi    Chennai
1          102  Priya  Coimbatore
2          103  Amit    Madurai
3          101  Ravi    Chennai
4          104  Neha      Salem
5          102  Priya  Coimbatore
6          105  Kiran    Trichy

Duplicate Records:

   CustomerID  Name      City
3          101  Ravi    Chennai
5          102  Priya  Coimbatore

Updated Dataset (Duplicates Removed):

   CustomerID  Name      City
0          101  Ravi    Chennai
1          102  Priya  Coimbatore
2          103  Amit    Madurai
4          104  Neha      Salem
6          105  Kiran    Trichy
|
```

## 7. Data Transformation

```
import pandas as pd

# Read the CSV file
df = pd.read_csv("products.csv")

# Display original dataset
print("Original Dataset:\n")
print(df)

# Conversion rate (1 USD = 83 INR)
exchange_rate = 83

# Add a new column 'Price_INR'
df["Price_INR"] = df["Price_USD"] * exchange_rate

# Display transformed dataset
print("\nTransformed Dataset:\n")
print(df)
```

```
===== RESTART: C:/Users/subha/Desktop/Query processing/11.py =====
Original Dataset:

   Product_ID Product_Name  Price_USD
0          101        Laptop         800
1          102         Mouse         20
2          103       Keyboard         35
3          104        Monitor        250
4          105   Headphones         50

Transformed Dataset:

   Product_ID Product_Name  Price_USD  Price_INR
0          101        Laptop         800     66400
1          102         Mouse         20      1660
2          103       Keyboard         35      2905
3          104        Monitor        250     20750
4          105   Headphones         50      4150
```

## 8. Data Aggregation

```
import pandas as pd

# Read the CSV file
df = pd.read_csv("monthly_sales.csv")

# Display original dataset
print("Monthly Sales Dataset:\n")
print(df)

# Calculate Total Sales
total_sales = df["Sales"].sum()

# Calculate Average Sales
average_sales = df["Sales"].mean()

# Calculate Maximum Sales
maximum_sales = df["Sales"].max()

# Calculate Minimum Sales
minimum_sales = df["Sales"].min()

# Display Results
print("\nSales Summary")
print("-----")
print("Total Sales   :", total_sales)
print("Average Sales :", average_sales)
print("Maximum Sales :", maximum_sales)
print("Minimum Sales :", minimum_sales)
```

```
===== RESTART: C:/Users/subha/Desktop/Query processing/sales.py =====
Monthly Sales Dataset:

   Month  Sales
0  January 25000
1  February 30000
2   March  28000
3   April  35000
4     May  32000
5    June  40000

Sales Summary
-----
Total Sales   : 190000
Average Sales : 31666.666666666668
Maximum Sales : 40000
Minimum Sales : 25000
|
```

## 9. Grouping Data

```
import pandas as pd

# Read the CSV file
df = pd.read_csv("employees.csv")

# Display original dataset
print("Employee Dataset:\n")
print(df)

# Group by Department
result = df.groupby("Department").agg(
    Total_Salary=("Salary", "sum"),
    Average_Salary=("Salary", "mean"),
    Employee_Count=("Salary", "count")
)

# Display grouped data
print("\nDepartment-wise Salary Summary:\n")
print(result)
```

```
===== RESTART: C:/Users/subha/Desktop/Query processing/employ.py =====
Employee Dataset:

   Employee_ID Employee_Name Department  Salary
0          101          Ravi         IT   50000
1          102          Priya         HR   45000
2          103           Amit         IT   60000
3          104           Neha    Finance   55000
4          105          Kiran         HR   48000
5          106           Arun    Finance   52000
6          107          Divya         IT   65000

Department-wise Salary Summary:

   Department  Total_Salary  Average_Salary  Employee_Count
Finance      107000      53500.000000           2
HR           93000      46500.000000           2
IT          175000      58333.333333           3
```

## 16. XML Data Analysis

```
import xml.etree.ElementTree as ET
```

```
# Read XML file
```

```
tree = ET.parse("library.xml")
```

```
root = tree.getroot()
```

```
# Total books
```

```
total_books = len(root.findall("book"))
```

```
print("Total Books:", total_books)
```

```
print("\nBooks Written After 2020:")
```

```
for book in root.findall("book"):
```

```
    year = int(book.find("year").text)
```

```
    if year > 2020:
```

```
        print(book.find("title").text)
```

```
print("\nBooks Costing More Than ₹500:")
```

```
for book in root.findall("book"):
```

```
    price = int(book.find("price").text)
```

```
    if price > 500:
```

```
        print(book.find("title").text)
```

```
===== RESTART: C:/Users/subha/Desktop/Query processing/books.py =====  
Total Books: 5  
  
Books Written After 2020:  
Data Science  
Machine Learning  
AI Essentials  
  
Books Costing More Than ₹500:  
Data Science  
Machine Learning  
AI Essentials
```

## 17. Data Interpretation using Python

```
# Store marks in a list
marks = [78, 45, 92, 36, 67, 81, 55, 39, 90, 48]
```

```
# Display marks
print("Student Marks:")
print(marks)
```

```
# Highest mark
highest = max(marks)
```

```
# Lowest mark
lowest = min(marks)
```

```
# Average mark
average = sum(marks) / len(marks)
```

```
# Number of students passed (Pass mark >= 40)
passed = 0
for mark in marks:
    if mark >= 40:
        passed += 1
```

```
# Display results
print("\nResults")
print("Highest Mark :", highest)
print("Lowest Mark :", lowest)
print("Average Mark :", round(average, 2))
print("Students Passed :", passed)
```

```
===== RESTART: C:/Users/subha/Desktop/Query processing/marks.py =====
Student Marks:
[78, 45, 92, 36, 67, 81, 55, 39, 90, 48]

Results
Highest Mark : 92
Lowest Mark  : 36
Average Mark : 63.1
Students Passed : 8
> |
```



## 18. Dictionary-Based Analysis

# Store employee information using dictionaries

```
employees = {  
    "Ravi": 50000,  
    "Priya": 45000,  
    "Amit": 60000,  
    "Neha": 55000,  
    "Karan": 48000  
}
```

# Highest salary employee

```
highest_employee = max(employees, key=employees.get)  
highest_salary = employees[highest_employee]
```

# Lowest salary employee

```
lowest_employee = min(employees, key=employees.get)  
lowest_salary = employees[lowest_employee]
```

# Average salary

```
average_salary = sum(employees.values()) / len(employees)
```

# Display results

```
print("Employee Details:")  
print(employees)
```

```
print("\nHighest Salary Employee:")  
print(highest_employee, "-", highest_salary)
```

```
print("\nLowest Salary Employee:")  
print(lowest_employee, "-", lowest_salary)
```

```
print("\nAverage Salary:")  
print(round(average_salary, 2))
```

```
===== RESTART: C:/Users/subha/Desktop/Query processing/dic.py =====  
Employee Details:  
{'Ravi': 50000, 'Priya': 45000, 'Amit': 60000, 'Neha': 55000, 'Karan': 48000}  
  
Highest Salary Employee:  
Amit - 60000  
  
Lowest Salary Employee:  
Priya - 45000  
  
Average Salary:  
51600.0
```